

Source folder: /Users/jonat/Documents/GitHub/master_thesis
Files in this part: 17

File: results_backup/dp_2_alpha_0.5_dim_64_clients_100_seed_123.json

```
-----
{
  "experiment_id": "dp_2_alpha_0.5_dim_64_clients_100_seed_123",
  "timestamp": "2026-02-11T02:42:57.478541",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 2,
    "use_dp": true,
    "dp_sigma": 40.7046875,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 123
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 3384.70252605272,
      "test_metrics": {
        "mse": 14.924642190551758,
        "mae": 3.540842860031128,
        "samples": 20000,
        "NDCG@10": 0.04847351432133537,
        "Hit@10": 0.07,
        "Precision@10": 0.007000000000000001,
        "Recall@10": 0.0016603465518112087,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 3507.607729213016,
      "test_metrics": {
        "mse": 18.18878537902832,
        "mae": 3.677388525772095,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 3,
      "train_loss": 3610.2767853764985,
      "test_metrics": {
        "mse": 23.594636837768554,
        "mae": 4.026535112762451,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 4,
      "train_loss": 3746.363727664809,
      "test_metrics": {
        "mse": 31.289648999023438,
```

```
"mae": 4.534889735412598,
"samples": 20000
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 5,
  "train_loss": 3856.7523849416866,
  "test_metrics": {
    "mse": 41.48843837890625,
    "mae": 5.180301279449463,
    "samples": 20000,
    "NDCG@10": 0.03870433732912934,
    "Hit@10": 0.03,
    "Precision@10": 0.0030000000000000005,
    "Recall@10": 0.002470653377630122,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 3951.9481274582445,
  "test_metrics": {
    "mse": 53.106045880126956,
    "mae": 5.82088099822998,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 4058.3124878553817,
  "test_metrics": {
    "mse": 67.12436173095703,
    "mae": 6.513197617340088,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 4151.770754174233,
  "test_metrics": {
    "mse": 83.58081507568359,
    "mae": 7.2610292114257815,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 4320.006720739383,
  "test_metrics": {
    "mse": 101.15629572753906,
    "mae": 8.025469793701172,
    "samples": 20000
  },
  "aggregation": {
```

```

        "num_clients": 100,
        "total_samples": 80000
    },
    {
        "round": 10,
        "train_loss": 4409.5620183469255,
        "test_metrics": {
            "mse": 122.77108743896484,
            "mae": 8.838040160369873,
            "samples": 20000,
            "NDCG@10": 0.03996836927225674,
            "Hit@10": 0.05,
            "Precision@10": 0.005,
            "Recall@10": 0.003423709984180309,
            "num_users_evaluated": 100
        },
        "aggregation": {
            "num_clients": 100,
            "total_samples": 80000
        }
    }
],
"final_metrics": {
    "mse": 122.77108743896484,
    "mae": 8.838040160369873,
    "samples": 20000,
    "NDCG@10": 0.03996836927225674,
    "Hit@10": 0.05,
    "Precision@10": 0.005,
    "Recall@10": 0.003423709984180309,
    "num_users_evaluated": 100
}
}

```

File: results_backup/dp_2_alpha_0.5_dim_64_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,3384.70252605272,0.04847351432133537,0.07,0.007000000000000001,0.0016603465518112087,14.924642190551758,3
.540842860031128
2,3507.607729213016,,,,,18.18878537902832,3.677388525772095
3,3610.2767853764985,,,,,23.594636837768554,4.026535112762451
4,3746.363727664809,,,,,31.289648999023438,4.534889735412598
5,3856.7523849416866,0.03870433732912934,0.03,0.0030000000000000005,0.002470653377630122,41.48843837890625,
5.180301279449463
6,3951.9481274582445,,,,,53.106045880126956,5.82088099822998
7,4058.3124878553817,,,,,67.12436173095703,6.513197617340088
8,4151.770754174233,,,,,83.58081507568359,7.2610292114257815
9,4320.006720739383,,,,,101.15629572753906,8.025469793701172
10,4409.5620183469255,0.03996836927225674,0.05,0.005,0.003423709984180309,122.77108743896484,8.838040160369
873
FINAL,,0.03996836927225674,0.05,0.005,0.003423709984180309,122.77108743896484,8.838040160369873

```

File: results_backup/dp_2_alpha_0.5_dim_64_clients_100_seed_42.json

```

{
  "experiment_id": "dp_2_alpha_0.5_dim_64_clients_100_seed_42",
  "timestamp": "2026-02-11T02:38:48.431009",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 2,
    "use_dp": true,
    "dp_sigma": 40.7046875,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
  },

```

```
"rounds": [
  {
    "round": 1,
    "train_loss": 3386.0751333259045,
    "test_metrics": {
      "mse": 14.867079779052734,
      "mae": 3.5347402950286866,
      "samples": 20000,
      "NDCG@10": 0.03971305784950692,
      "Hit@10": 0.06,
      "Precision@10": 0.006999999999999999,
      "Recall@10": 0.0027202326042891443,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 2,
    "train_loss": 3488.49369102608,
    "test_metrics": {
      "mse": 17.96334690246582,
      "mae": 3.6511829357147216,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 3,
    "train_loss": 3620.4820652038243,
    "test_metrics": {
      "mse": 23.45289249267578,
      "mae": 4.01370072479248,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 4,
    "train_loss": 3707.3488921543467,
    "test_metrics": {
      "mse": 31.128795666503905,
      "mae": 4.532878741455078,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 5,
    "train_loss": 3833.9187078650975,
    "test_metrics": {
      "mse": 40.42511262817383,
      "mae": 5.121992610931397,
      "samples": 20000,
      "NDCG@10": 0.04759520380012195,
      "Hit@10": 0.08,
      "Precision@10": 0.009000000000000001,
      "Recall@10": 0.005139105992047169,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
]
```

```
}
},
{
  "round": 6,
  "train_loss": 3908.0879557490157,
  "test_metrics": {
    "mse": 51.90749056396484,
    "mae": 5.775124337005615,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 4082.852249748841,
  "test_metrics": {
    "mse": 66.30026739501953,
    "mae": 6.4860578056335445,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 4174.175201685227,
  "test_metrics": {
    "mse": 80.15156484375,
    "mae": 7.126118628692627,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 4266.877559568634,
  "test_metrics": {
    "mse": 98.07662218017578,
    "mae": 7.861813136291504,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 4389.6386369329675,
  "test_metrics": {
    "mse": 118.63443210449219,
    "mae": 8.673735711669922,
    "samples": 20000,
    "NDCG@10": 0.05896409118693234,
    "Hit@10": 0.07,
    "Precision@10": 0.0070000000000000001,
    "Recall@10": 0.0021049281556513627,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
```

```

    "mse": 118.63443210449219,
    "mae": 8.673735711669922,
    "samples": 20000,
    "NDCG@10": 0.05896409118693234,
    "Hit@10": 0.07,
    "Precision@10": 0.007000000000000001,
    "Recall@10": 0.0021049281556513627,
    "num_users_evaluated": 100
  },
  "attack_results": {
    "mia": {
      "train_accuracy": 1.0,
      "auc": 0.5004,
      "accuracy": 0.5,
      "true_positive_rate": 0.64,
      "false_positive_rate": 0.64
    },
    "inversion": {
      "top_k_accuracy": 0.0,
      "top_k": 10,
      "num_users_evaluated": 50
    }
  }
}

```

File: results_backup/dp_2_alpha_0.5_dim_64_clients_100_seed_42_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,3386.0751333259045,0.03971305784950692,0.06,0.006999999999999999,0.0027202326042891443,14.867079779052734
,3.5347402950286866
2,3488.49369102608,,,,,17.96334690246582,3.6511829357147216
3,3620.4820652038243,,,,,23.45289249267578,4.01370072479248
4,3707.3488921543467,,,,,31.128795666503905,4.532878741455078
5,3833.9187078650975,0.04759520380012195,0.08,0.009000000000000001,0.005139105992047169,40.42511262817383,5
.121992610931397
6,3908.0879557490157,,,,,51.90749056396484,5.775124337005615
7,4082.852249748841,,,,,66.30026739501953,6.4860578056335445
8,4174.175201685227,,,,,80.15156484375,7.126118628692627
9,4266.877559568634,,,,,98.07662218017578,7.861813136291504
10,4389.6386369329675,0.05896409118693234,0.07,0.007000000000000001,0.0021049281556513627,118.6344321044921
9,8.673735711669922
FINAL,,0.05896409118693234,0.07,0.007000000000000001,0.0021049281556513627,118.63443210449219,8.67373571166
9922

```

File: results_backup/dp_2_alpha_0.5_dim_64_clients_100_seed_456.json

```

{
  "experiment_id": "dp_2_alpha_0.5_dim_64_clients_100_seed_456",
  "timestamp": "2026-02-11T02:46:22.816395",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 2,
    "use_dp": true,
    "dp_sigma": 40.7046875,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 3379.9243578834735,
      "test_metrics": {
        "mse": 14.82579482421875,
        "mae": 3.5305438163757326,
        "samples": 20000,
        "NDCG@10": 0.04543685807395247,

```

```
"Hit@10": 0.09,
"Precision@10": 0.009000000000000001,
"Recall@10": 0.003351877595037266,
"num_users_evaluated": 100
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 2,
  "train_loss": 3499.3596683412675,
  "test_metrics": {
    "mse": 18.088604051208495,
    "mae": 3.6784022804260252,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 3,
  "train_loss": 3579.559214961313,
  "test_metrics": {
    "mse": 23.290728747558592,
    "mae": 4.014856020355225,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 3695.6306296415255,
  "test_metrics": {
    "mse": 30.755915740966795,
    "mae": 4.5036157135009764,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 3817.37629137021,
  "test_metrics": {
    "mse": 40.1239346496582,
    "mae": 5.095039193725586,
    "samples": 20000,
    "NDCG@10": 0.04656793552925941,
    "Hit@10": 0.05,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.0026509588557606283,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 3917.4636466761917,
  "test_metrics": {
    "mse": 51.799483715820315,
    "mae": 5.744916427612305,
    "samples": 20000
```

```
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 7,
    "train_loss": 4077.879251762293,
    "test_metrics": {
      "mse": 65.4899167602539,
      "mae": 6.441198127746582,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 8,
    "train_loss": 4145.120165642374,
    "test_metrics": {
      "mse": 81.12905133056641,
      "mae": 7.158955921173096,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 9,
    "train_loss": 4258.075307701938,
    "test_metrics": {
      "mse": 97.96482468261719,
      "mae": 7.883077192687988,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 10,
    "train_loss": 4376.930281931637,
    "test_metrics": {
      "mse": 117.72294920654296,
      "mae": 8.617804049682617,
      "samples": 20000,
      "NDCG@10": 0.03776224559152501,
      "Hit@10": 0.05,
      "Precision@10": 0.005,
      "Recall@10": 0.0023605155007594032,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
],
"final_metrics": {
  "mse": 117.72294920654296,
  "mae": 8.617804049682617,
  "samples": 20000,
  "NDCG@10": 0.03776224559152501,
  "Hit@10": 0.05,
  "Precision@10": 0.005,
  "Recall@10": 0.0023605155007594032,
  "num_users_evaluated": 100
}
```



```
}
```

File: results_backup/dp_2_alpha_0.5_dim_64_clients_100_seed_456_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,3379.9243578834735,0.04543685807395247,0.09,0.009000000000000001,0.003351877595037266,14.82579482421875,3
.5305438163757326
2,3499.3596683412675,,,,,18.088604051208495,3.6784022804260252
3,3579.559214961313,,,,,23.290728747558592,4.014856020355225
4,3695.6306296415255,,,,,30.755915740966795,4.5036157135009764
5,3817.37629137021,0.04656793552925941,0.05,0.006000000000000001,0.0026509588557606283,40.1239346496582,5.0
95039193725586
6,3917.4636466761917,,,,,51.799483715820315,5.744916427612305
7,4077.879251762293,,,,,65.4899167602539,6.441198127746582
8,4145.120165642374,,,,,81.12905133056641,7.158955921173096
9,4258.075307701938,,,,,97.96482468261719,7.883077192687988
10,4376.930281931637,0.03776224559152501,0.05,0.005,0.0023605155007594032,117.72294920654296,8.617804049682
617
FINAL,,0.03776224559152501,0.05,0.005,0.0023605155007594032,117.72294920654296,8.617804049682617
```

File: results_backup/dp_4_alpha_0.5_dim_64_clients_100_seed_123.json

```
{
  "experiment_id": "dp_4_alpha_0.5_dim_64_clients_100_seed_123",
  "timestamp": "2026-02-11T02:31:10.230233",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 4,
    "use_dp": true,
    "dp_sigma": 20.40234375,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 123
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 227.39276497234422,
      "test_metrics": {
        "mse": 13.800793028259278,
        "mae": 3.522211731719971,
        "samples": 20000,
        "NDCG@10": 0.05185266953566699,
        "Hit@10": 0.06,
        "Precision@10": 0.006000000000000001,
        "Recall@10": 0.0011885987128906943,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 235.41270588546533,
      "test_metrics": {
        "mse": 14.025225309753418,
        "mae": 3.5217144886016847,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    }
  ],
  {
```

```
"round": 3,
"train_loss": 241.86451076223912,
"test_metrics": {
  "mse": 14.380878532409668,
  "mae": 3.52511671333313,
  "samples": 20000
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 4,
  "train_loss": 250.65472324378024,
  "test_metrics": {
    "mse": 14.879275038146973,
    "mae": 3.532377660369873,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 258.0789284126659,
  "test_metrics": {
    "mse": 15.601707807922363,
    "mae": 3.5585600902557375,
    "samples": 20000,
    "NDCG@10": 0.043454917351544095,
    "Hit@10": 0.05,
    "Precision@10": 0.005,
    "Recall@10": 0.0038039867109634553,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 263.62437503811003,
  "test_metrics": {
    "mse": 16.389767547607423,
    "mae": 3.5958754760742186,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 270.9868775376554,
  "test_metrics": {
    "mse": 17.26231935119629,
    "mae": 3.630116690826416,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 276.4068795810124,
  "test_metrics": {
    "mse": 18.307551095581054,
```

```

    "mae": 3.6863674404144287,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 286.78034017667227,
  "test_metrics": {
    "mse": 19.379044264221193,
    "mae": 3.753527548599243,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 292.1784286954096,
  "test_metrics": {
    "mse": 20.81909543457031,
    "mae": 3.850083127212524,
    "samples": 20000,
    "NDCG@10": 0.048026363096703564,
    "Hit@10": 0.06,
    "Precision@10": 0.006,
    "Recall@10": 0.00366180522275547,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
  "mse": 20.81909543457031,
  "mae": 3.850083127212524,
  "samples": 20000,
  "NDCG@10": 0.048026363096703564,
  "Hit@10": 0.06,
  "Precision@10": 0.006,
  "Recall@10": 0.00366180522275547,
  "num_users_evaluated": 100
}
}

```

File: results_backup/dp_4_alpha_0.5_dim_64_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,227.39276497234422,0.05185266953566699,0.06,0.006000000000000001,0.0011885987128906943,13.800793028259278
,3.522211731719971
2,235.41270588546533,,,,,14.025225309753418,3.5217144886016847
3,241.86451076223912,,,,,14.380878532409668,3.52511671333313
4,250.65472324378024,,,,,14.879275038146973,3.532377660369873
5,258.0789284126659,0.043454917351544095,0.05,0.005,0.0038039867109634553,15.601707807922363,3.558560090255
7375
6,263.62437503811003,,,,,16.389767547607423,3.5958754760742186
7,270.9868775376554,,,,,17.26231935119629,3.630116690826416
8,276.4068795810124,,,,,18.307551095581054,3.6863674404144287
9,286.78034017667227,,,,,19.379044264221193,3.753527548599243
10,292.1784286954096,0.048026363096703564,0.06,0.006,0.00366180522275547,20.81909543457031,3.8500831272125
24
FINAL,,0.048026363096703564,0.06,0.006,0.00366180522275547,20.81909543457031,3.850083127212524

```

File: results_backup/dp_4_alpha_0.5_dim_64_clients_100_seed_42.json

```

{
  "experiment_id": "dp_4_alpha_0.5_dim_64_clients_100_seed_42",

```

```
"timestamp": "2026-02-11T02:26:43.406751",
"config": {
  "num_users": 943,
  "num_items": 1682,
  "embedding_dim": 64,
  "num_clients": 100,
  "alpha": 0.5,
  "dp_epsilon": 4,
  "use_dp": true,
  "dp_sigma": 20.40234375,
  "dp_clip_norm": 1.0,
  "num_rounds": 10,
  "local_epochs": 3,
  "learning_rate": 0.01,
  "batch_size": 32,
  "seed": 42
},
"rounds": [
  {
    "round": 1,
    "train_loss": 228.00020887873922,
    "test_metrics": {
      "mse": 13.7829908203125,
      "mae": 3.5202133369445803,
      "samples": 20000,
      "NDCG@10": 0.06573527998901081,
      "Hit@10": 0.11,
      "Precision@10": 0.011000000000000001,
      "Recall@10": 0.005121197136832625,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 2,
    "train_loss": 234.42883935513137,
    "test_metrics": {
      "mse": 13.968070623779298,
      "mae": 3.513886647796631,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 3,
    "train_loss": 242.85100448068195,
    "test_metrics": {
      "mse": 14.35976184539795,
      "mae": 3.523405859375,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 4,
    "train_loss": 248.08766878346557,
    "test_metrics": {
      "mse": 14.908065705871582,
      "mae": 3.540966688156128,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
]
```

```
},
{
  "round": 5,
  "train_loss": 256.18368029690765,
  "test_metrics": {
    "mse": 15.50926480102539,
    "mae": 3.5578789264678954,
    "samples": 20000,
    "NDCG@10": 0.04759706688923452,
    "Hit@10": 0.07,
    "Precision@10": 0.008,
    "Recall@10": 0.00414447614931606,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 260.72376313686743,
  "test_metrics": {
    "mse": 16.2442454788208,
    "mae": 3.5853822803497315,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 271.36841570603195,
  "test_metrics": {
    "mse": 17.139047613525392,
    "mae": 3.619733070373535,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 277.6128170244431,
  "test_metrics": {
    "mse": 17.961268449401857,
    "mae": 3.6585649604797363,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 283.44371966813014,
  "test_metrics": {
    "mse": 19.153935430908202,
    "mae": 3.7339311351776123,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 291.8478941332176,
```

```

    "test_metrics": {
      "mse": 20.485859423828124,
      "mae": 3.8241310848236085,
      "samples": 20000,
      "NDCG@10": 0.05901090278606798,
      "Hit@10": 0.07,
      "Precision@10": 0.006999999999999999,
      "Recall@10": 0.0021080536440089186,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  "final_metrics": {
    "mse": 20.485859423828124,
    "mae": 3.8241310848236085,
    "samples": 20000,
    "NDCG@10": 0.05901090278606798,
    "Hit@10": 0.07,
    "Precision@10": 0.006999999999999999,
    "Recall@10": 0.0021080536440089186,
    "num_users_evaluated": 100
  },
  "attack_results": {
    "mia": {
      "train_accuracy": 1.0,
      "auc": 0.5024500000000001,
      "accuracy": 0.505,
      "true_positive_rate": 0.61,
      "false_positive_rate": 0.6
    },
    "inversion": {
      "top_k_accuracy": 0.0,
      "top_k": 10,
      "num_users_evaluated": 50
    }
  }
}

```

File: results_backup/dp_4_alpha_0.5_dim_64_clients_100_seed_42_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,228.00020887873922,0.06573527998901081,0.11,0.011000000000000001,0.005121197136832625,13.7829908203125,3.5202133369445803
2,234.42883935513137,,,,,13.968070623779298,3.513886647796631
3,242.85100448068195,,,,,14.35976184539795,3.523405859375
4,248.08766878346557,,,,,14.908065705871582,3.540966688156128
5,256.18368029690765,0.04759706688923452,0.07,0.008,0.00414447614931606,15.50926480102539,3.557878926467895
6,260.72376313686743,,,,,16.2442454788208,3.5853822803497315
7,271.36841570603195,,,,,17.139047613525392,3.619733070373535
8,277.6128170244431,,,,,17.961268449401857,3.6585649604797363
9,283.44371966813014,,,,,19.153935430908202,3.7339311351776123
10,291.8478941332176,0.05901090278606798,0.07,0.006999999999999999,0.0021080536440089186,20.485859423828124,3.8241310848236085
FINAL,,0.05901090278606798,0.07,0.006999999999999999,0.0021080536440089186,20.485859423828124,3.8241310848236085

```

File: results_backup/dp_4_alpha_0.5_dim_64_clients_100_seed_456.json

```

{
  "experiment_id": "dp_4_alpha_0.5_dim_64_clients_100_seed_456",
  "timestamp": "2026-02-11T02:34:47.725248",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 4,
    "use_dp": true,

```

```
"dp_sigma": 20.40234375,
"dp_clip_norm": 1.0,
"num_rounds": 10,
"local_epochs": 3,
"learning_rate": 0.01,
"batch_size": 32,
"seed": 456
},
"rounds": [
  {
    "round": 1,
    "train_loss": 227.25744999731683,
    "test_metrics": {
      "mse": 13.783183213806153,
      "mae": 3.5202975929260254,
      "samples": 20000,
      "NDCG@10": 0.058843959939489814,
      "Hit@10": 0.11,
      "Precision@10": 0.011000000000000001,
      "Recall@10": 0.0043079012461828695,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 2,
    "train_loss": 234.6616195220771,
    "test_metrics": {
      "mse": 14.0265330078125,
      "mae": 3.5235395816802977,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 3,
    "train_loss": 240.2358970889224,
    "test_metrics": {
      "mse": 14.386404266357422,
      "mae": 3.529303728866577,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 4,
    "train_loss": 247.30434022342914,
    "test_metrics": {
      "mse": 14.87741611480713,
      "mae": 3.538506480407715,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 5,
    "train_loss": 254.62335761036957,
    "test_metrics": {
      "mse": 15.513196069335937,
      "mae": 3.5569311714172365,
      "samples": 20000,
      "NDCG@10": 0.04920733657218897,
```

```
"Hit@10": 0.05,
"Precision@10": 0.005,
"Recall@10": 0.0025243765772796157,
"num_users_evaluated": 100
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 6,
  "train_loss": 261.4700665301408,
  "test_metrics": {
    "mse": 16.287284356689455,
    "mae": 3.5860441093444826,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 271.48093422458817,
  "test_metrics": {
    "mse": 17.255915408325194,
    "mae": 3.636420043182373,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 276.1572551277892,
  "test_metrics": {
    "mse": 18.261986018371584,
    "mae": 3.688884970855713,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 282.6003147611433,
  "test_metrics": {
    "mse": 19.30410701751709,
    "mae": 3.746032738494873,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 290.2898528725529,
  "test_metrics": {
    "mse": 20.64410247039795,
    "mae": 3.8355876289367674,
    "samples": 20000,
    "NDCG@10": 0.03669679703804605,
    "Hit@10": 0.05,
    "Precision@10": 0.005,
    "Recall@10": 0.0022616144018583043,
    "num_users_evaluated": 100
  }
}
```



```

    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
},
"final_metrics": {
  "mse": 20.64410247039795,
  "mae": 3.8355876289367674,
  "samples": 20000,
  "NDCG@10": 0.03669679703804605,
  "Hit@10": 0.05,
  "Precision@10": 0.005,
  "Recall@10": 0.0022616144018583043,
  "num_users_evaluated": 100
}
}

```

File: results_backup/dp_4_alpha_0.5_dim_64_clients_100_seed_456_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,227.25744999731683,0.058843959939489814,0.11,0.011000000000000001,0.0043079012461828695,13.78318321380615
3,3.5202975929260254
2,234.6616195220771,,,,,14.0265330078125,3.5235395816802977
3,240.2358970889224,,,,,14.386404266357422,3.529303728866577
4,247.30434022342914,,,,,14.87741611480713,3.538506480407715
5,254.62335761036957,0.04920733657218897,0.05,0.005,0.0025243765772796157,15.513196069335937,3.556931171417
2365
6,261.4700665301408,,,,,16.287284356689455,3.5860441093444826
7,271.48093422458817,,,,,17.255915408325194,3.636420043182373
8,276.1572551277892,,,,,18.261986018371584,3.688884970855713
9,282.60031476114433,,,,,19.30410701751709,3.746032738494873
10,290.2898528725529,0.03669679703804605,0.05,0.005,0.0022616144018583043,20.64410247039795,3.8355876289367
674
FINAL,,0.03669679703804605,0.05,0.005,0.0022616144018583043,20.64410247039795,3.8355876289367674

```

File: results_backup/dp_8_alpha_0.5_dim_64_clients_100_seed_123.json

```

{
  "experiment_id": "dp_8_alpha_0.5_dim_64_clients_100_seed_123",
  "timestamp": "2026-02-11T02:17:59.276389",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 8,
    "use_dp": true,
    "dp_sigma": 11.03203125,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 123
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 32.2485865604616,
      "test_metrics": {
        "mse": 13.706314096069336,
        "mae": 3.5218424324035644,
        "samples": 20000,
        "NDCG@10": 0.056472994498314326,
        "Hit@10": 0.06,
        "Precision@10": 0.006000000000000001,
        "Recall@10": 0.0016320618662437314,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,

```

```
    "total_samples": 80000
  },
  {
    "round": 2,
    "train_loss": 33.034346703759404,
    "test_metrics": {
      "mse": 13.729706134033203,
      "mae": 3.5210181625366213,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 3,
    "train_loss": 33.56999220028218,
    "test_metrics": {
      "mse": 13.764635670471192,
      "mae": 3.520541397857666,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 4,
    "train_loss": 34.37596866098962,
    "test_metrics": {
      "mse": 13.807001486206055,
      "mae": 3.519665375137329,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 5,
    "train_loss": 35.117718829367234,
    "test_metrics": {
      "mse": 13.88942387084961,
      "mae": 3.521537496948242,
      "samples": 20000,
      "NDCG@10": 0.04564447310757874,
      "Hit@10": 0.08,
      "Precision@10": 0.008,
      "Recall@10": 0.005146976538473407,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 6,
    "train_loss": 35.47923740492363,
    "test_metrics": {
      "mse": 13.97281569519043,
      "mae": 3.5236828666687012,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
```

```

    "round": 7,
    "train_loss": 36.279913027801626,
    "test_metrics": {
      "mse": 14.046265171813966,
      "mae": 3.5218750690460205,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 8,
    "train_loss": 36.62263538561832,
    "test_metrics": {
      "mse": 14.13695319519043,
      "mae": 3.5216548866271973,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 9,
    "train_loss": 37.43999281625264,
    "test_metrics": {
      "mse": 14.212836094665528,
      "mae": 3.5201156791687014,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 10,
    "train_loss": 37.8632549337846,
    "test_metrics": {
      "mse": 14.355028085327149,
      "mae": 3.524582919692993,
      "samples": 20000,
      "NDCG@10": 0.05788043055404635,
      "Hit@10": 0.07,
      "Precision@10": 0.006999999999999999,
      "Recall@10": 0.004431035991506316,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
],
"final_metrics": {
  "mse": 14.355028085327149,
  "mae": 3.524582919692993,
  "samples": 20000,
  "NDCG@10": 0.05788043055404635,
  "Hit@10": 0.07,
  "Precision@10": 0.006999999999999999,
  "Recall@10": 0.004431035991506316,
  "num_users_evaluated": 100
}
}

```

File: results_backup/dp_8_alpha_0.5_dim_64_clients_100_seed_123_summary.csv

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
 1,32.2485865604616,0.056472994498314326,0.06,0.006000000000000001,0.0016320618662437314,13.706314096069336,
 3.5218424324035644

```
2,33.034346703759404,,,,,13.729706134033203,3.5210181625366213
3,33.56999220028218,,,,,13.764635670471192,3.520541397857666
4,34.37596866098962,,,,,13.807001486206055,3.519665375137329
5,35.117718829367234,0.04564447310757874,0.08,0.008,0.005146976538473407,13.88942387084961,3.52153749694824
2
6,35.47923740492363,,,,,13.97281569519043,3.5236828666687012
7,36.279913027801626,,,,,14.046265171813966,3.5218750690460205
8,36.62263538561832,,,,,14.13695319519043,3.5216548866271973
9,37.43999281625264,,,,,14.212836094665528,3.5201156791687014
10,37.8632549337846,0.05788043055404635,0.07,0.006999999999999999,0.004431035991506316,14.355028085327149,3
.524582919692993
FINAL,,0.05788043055404635,0.07,0.006999999999999999,0.004431035991506316,14.355028085327149,3.524582919692
993
```

File: results_backup/dp_8_alpha_0.5_dim_64_clients_100_seed_42.json

```
{
  "experiment_id": "dp_8_alpha_0.5_dim_64_clients_100_seed_42",
  "timestamp": "2026-02-11T02:14:30.202965",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 8,
    "use_dp": true,
    "dp_sigma": 11.03203125,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 32.46484567453981,
      "test_metrics": {
        "mse": 13.70007797088623,
        "mae": 3.521318423843384,
        "samples": 20000,
        "NDCG@10": 0.07062999616437177,
        "Hit@10": 0.1,
        "Precision@10": 0.01,
        "Recall@10": 0.004255373351798951,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 32.984528741289346,
      "test_metrics": {
        "mse": 13.713029345703125,
        "mae": 3.5190051342010498,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 3,
      "train_loss": 33.73158305571247,
      "test_metrics": {
        "mse": 13.760207579040527,
        "mae": 3.520247960281372,
        "samples": 20000
      }
    }
  ]
}
```

```
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 4,
    "train_loss": 34.126739832006564,
    "test_metrics": {
      "mse": 13.82516392211914,
      "mae": 3.522406580734253,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 5,
    "train_loss": 34.842418864136576,
    "test_metrics": {
      "mse": 13.877694001770019,
      "mae": 3.5213635154724123,
      "samples": 20000,
      "NDCG@10": 0.038458220330608306,
      "Hit@10": 0.07,
      "Precision@10": 0.0070000000000000001,
      "Recall@10": 0.0038556439145193927,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 6,
    "train_loss": 35.2034369765747,
    "test_metrics": {
      "mse": 13.941607377624512,
      "mae": 3.520322141265869,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 7,
    "train_loss": 36.004213745426235,
    "test_metrics": {
      "mse": 14.01173528137207,
      "mae": 3.5188413883209226,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 8,
    "train_loss": 36.656923031942206,
    "test_metrics": {
      "mse": 14.06484306640625,
      "mae": 3.5142020988464355,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
```

```

    }
  },
  {
    "round": 9,
    "train_loss": 37.13889551900223,
    "test_metrics": {
      "mse": 14.185741499328614,
      "mae": 3.517760985183716,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 10,
    "train_loss": 38.064433839840845,
    "test_metrics": {
      "mse": 14.309214770507813,
      "mae": 3.5198187145233155,
      "samples": 20000,
      "NDCG@10": 0.06680998371750531,
      "Hit@10": 0.06,
      "Precision@10": 0.006,
      "Recall@10": 0.0017234382593935339,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
],
"final_metrics": {
  "mse": 14.309214770507813,
  "mae": 3.5198187145233155,
  "samples": 20000,
  "NDCG@10": 0.06680998371750531,
  "Hit@10": 0.06,
  "Precision@10": 0.006,
  "Recall@10": 0.0017234382593935339,
  "num_users_evaluated": 100
},
"attack_results": {
  "mia": {
    "train_accuracy": 1.0,
    "auc": 0.53355,
    "accuracy": 0.515,
    "true_positive_rate": 0.6,
    "false_positive_rate": 0.57
  },
  "inversion": {
    "top_k_accuracy": 0.0,
    "top_k": 10,
    "num_users_evaluated": 50
  }
}
}

```

File: results_backup/dp_8_alpha_0.5_dim_64_clients_100_seed_42_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,32.46484567453981,0.07062999616437177,0.1,0.01,0.004255373351798951,13.70007797088623,3.521318423843384
2,32.984528741289346,,,,,13.713029345703125,3.5190051342010498
3,33.73158305571247,,,,,13.760207579040527,3.520247960281372
4,34.126739832006564,,,,,13.82516392211914,3.522406580734253
5,34.842418864136576,0.038458220330608306,0.07,0.007000000000000001,0.0038556439145193927,13.87769400177001
9,3.5213635154724123
6,35.2034369765747,,,,,13.941607377624512,3.520322141265869
7,36.004213745426235,,,,,14.01173528137207,3.5188413883209226
8,36.656923031942206,,,,,14.06484306640625,3.5142020988464355
9,37.13889551900223,,,,,14.185741499328614,3.517760985183716
10,38.064433839840845,0.06680998371750531,0.06,0.006,0.0017234382593935339,14.309214770507813,3.51981871452

```

33155

FINAL,,0.06680998371750531,0.06,0.006,0.0017234382593935339,14.309214770507813,3.5198187145233155

File: results_backup/dp_8_alpha_0.5_dim_64_clients_100_seed_456.json

```
-----
{
  "experiment_id": "dp_8_alpha_0.5_dim_64_clients_100_seed_456",
  "timestamp": "2026-02-11T02:22:26.614168",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 8,
    "use_dp": true,
    "dp_sigma": 11.03203125,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 32.29933751890424,
      "test_metrics": {
        "mse": 13.70424580230713,
        "mae": 3.5219111457824708,
        "samples": 20000,
        "NDCG@10": 0.04900664415170795,
        "Hit@10": 0.07,
        "Precision@10": 0.007000000000000001,
        "Recall@10": 0.0020227477003257922,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 32.88257406573976,
      "test_metrics": {
        "mse": 13.735208779907227,
        "mae": 3.522513412475586,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 3,
      "train_loss": 33.490608923782005,
      "test_metrics": {
        "mse": 13.773474458312988,
        "mae": 3.52244154586792,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 4,
      "train_loss": 34.03537829007678,
      "test_metrics": {
        "mse": 13.8231083984375,
```

```
"mae": 3.5225142669677734,
"samples": 20000
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 5,
  "train_loss": 34.5790928287218,
  "test_metrics": {
    "mse": 13.886728021240234,
    "mae": 3.522496696472168,
    "samples": 20000,
    "NDCG@10": 0.06690983654996545,
    "Hit@10": 0.09,
    "Precision@10": 0.011000000000000001,
    "Recall@10": 0.007239654355057395,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 35.31948951689446,
  "test_metrics": {
    "mse": 13.964869668579102,
    "mae": 3.5228670364379884,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 36.126836900261964,
  "test_metrics": {
    "mse": 14.081374354553223,
    "mae": 3.5267831939697265,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 36.630888192507676,
  "test_metrics": {
    "mse": 14.17117515258789,
    "mae": 3.5257721771240234,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 36.98664217515131,
  "test_metrics": {
    "mse": 14.257159707641602,
    "mae": 3.5239249969482422,
    "samples": 20000
  },
  "aggregation": {
```



```
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 37.696265293993996,
  "test_metrics": {
    "mse": 14.396677098083495,
    "mae": 3.529047723388672,
    "samples": 20000,
    "NDCG@10": 0.03561013735332017,
    "Hit@10": 0.05,
    "Precision@10": 0.005,
    "Recall@10": 0.001212587264121489,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
  "mse": 14.396677098083495,
  "mae": 3.529047723388672,
  "samples": 20000,
  "NDCG@10": 0.03561013735332017,
  "Hit@10": 0.05,
  "Precision@10": 0.005,
  "Recall@10": 0.001212587264121489,
  "num_users_evaluated": 100
}
}
```